Group 3: Dog Treat Dispenser Spur Gears

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Table of Roles

Jessie Zhang	Design sketch, annotation, research, dimensions, mechanical interfaces
Raffaela Mase	Design sketch, annotation, research, dimensions, mechanical interfaces
Matthew Kuan	Design sketch, annotation, research, dimensions, mechanical interfaces
Xiaoting Liu	Design sketch, annotation, research, dimensions, mechanical interfaces
Nipon Sajib	Design sketch, annotation, research, dimensions, mechanical interfaces

Subteams: assembly drawings, orthographic projections (MK), and motion simulations.

Idea 1 : Jessie Zhang

1. **Geometric Constraints**
   - ***Maximum Outer Diameter (OD)***: ≤ 4.65 in. (ensures spatial clearance)
   - ***Gear Thickness***: 1.691 in. (guarantees axial alignment)
   - ***Drive Interface:*** 0.383 in. × 0.383 in. centered square bore to transmit torque w/out slipping

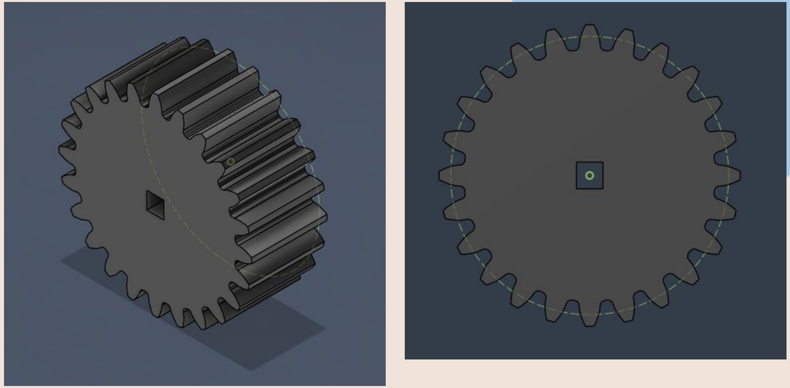
2. **Kinematic Calculations** [4]
   - Physical treat diameter **(0.25 in.)** dictates the required tooth spacing along Pitch Diameter (PD) ≅ 0.262 in.
   - ***Calculated PD*** = 6 (allows smooth mechanical clearance)
   - ***Determined # of Teeth***: N = 24

     Using: $N = 2PD/(OD - PD)$

     Verification: $DP = (N + 2)/OD$

     ✅ **Resulting OD** = 4.333 in. satisfies constraints

3. **Software Implementation**
   - Generated via Fusion 360 Spur Gear tool
     - ❑ English units, 20 deg. pressure angle
   - Executed an extrude cut to create the center square connector.

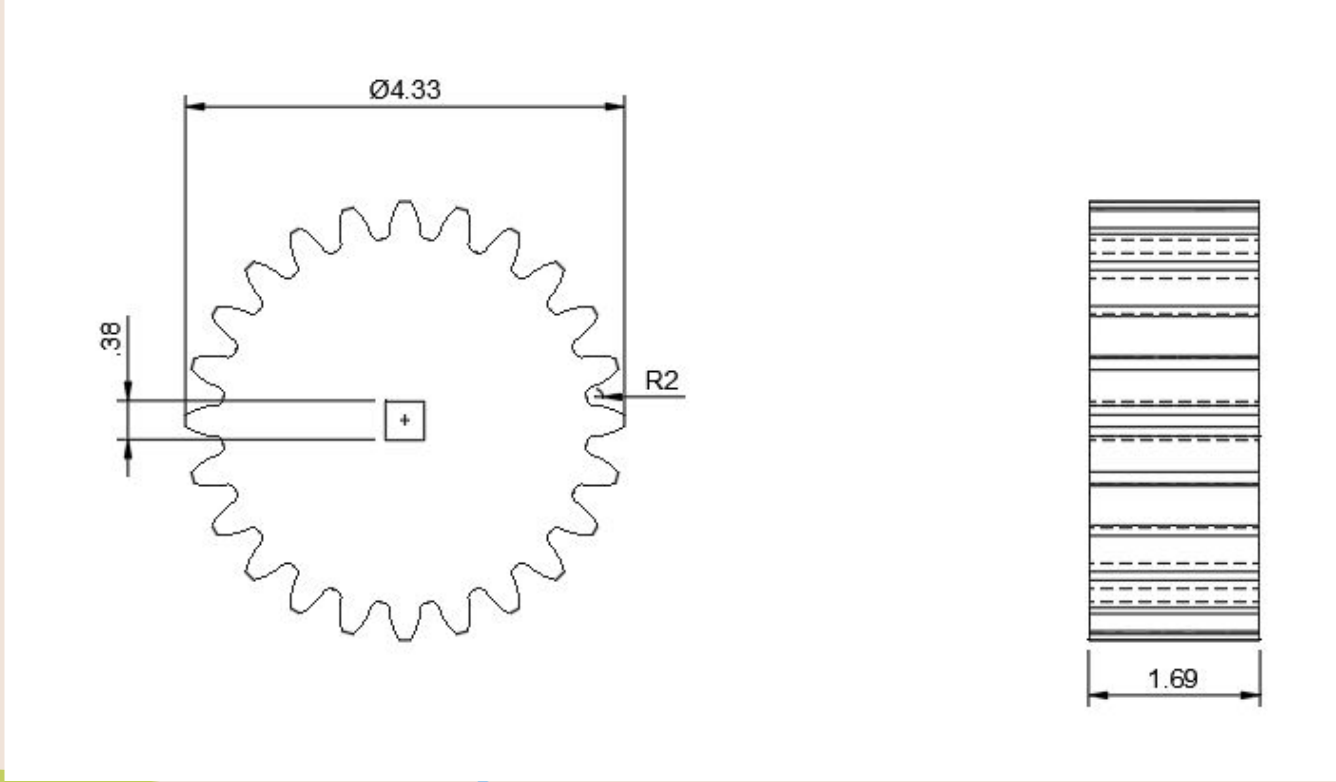


Figures 1 & 2: Sketch generated using Fusion 260 Spur Gear Tool

Figure 2: Dog treats used for reference to determine PD [5].

Idea 1 : Jessie Zhang

4. Annotated Sketch



Xiaoting Liu

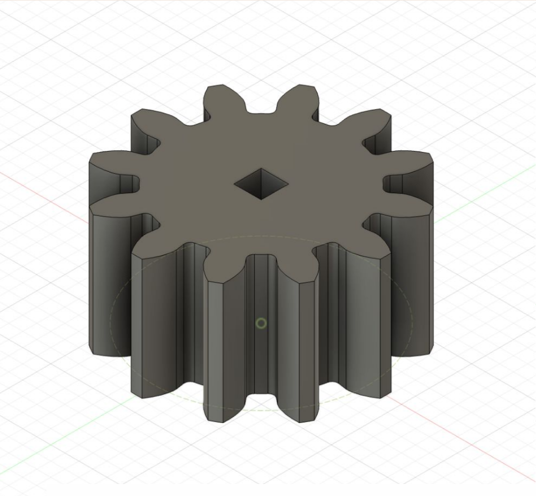
Idea 2

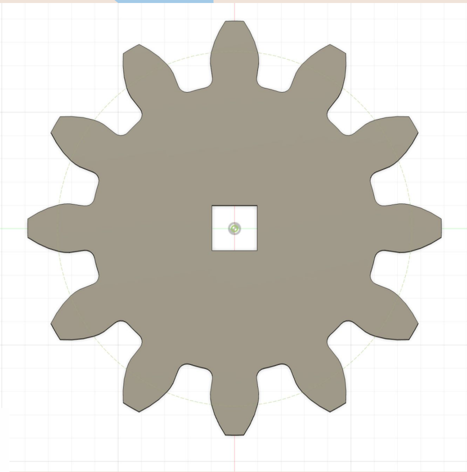
Chosen parameters:

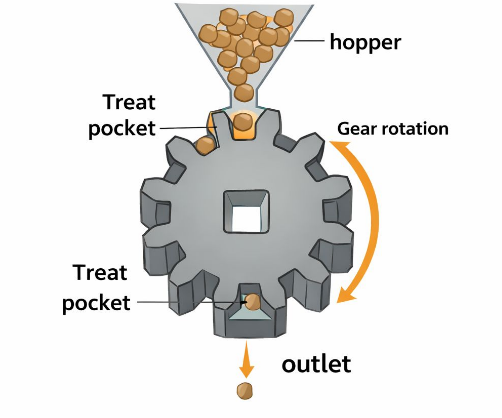
- Standard: English, In unit
- Pressure Angle: 20 degree
- Backlash: 0.00 in
- Root fillet radius: 0.079 in
- Number of teeth (N): 12
  N=2PD/OD-PD = 2(3)/(3.5-3) = 12
- Outer diameter (OD): 3.5 in
- Diametral pitch (DP): 4
  DP = N+2/OD = (12+2)/3.5 4
- Pitch Diameter (PD): 3 in
- Gear thickness :1.691in
- Dimension of the square hole: 0.383 in x 0.383 in
- Recommended treat diameter: ≤ 0.35 in

Idea for ensuring only one treat at a time

To have a small hopper on top of the gear, where the end of the hopper is just big enough for one treat to go through. This ensures only one treat on each teeth of the gear.







Each treat is stored in an individual pocket between gear teeth

Idea 3: Matthew Kuan

Justification for the chosen design based on user needs or technical constraints.

- Dispensing gear stores treats between gaps in teeth
- Inner gear rotates, driving the outer gear to drop one treat at a time
- Ratcheting mechanism and recoil spring [2]
  - Allows lever to spring back for user accessibility without dispensing treat

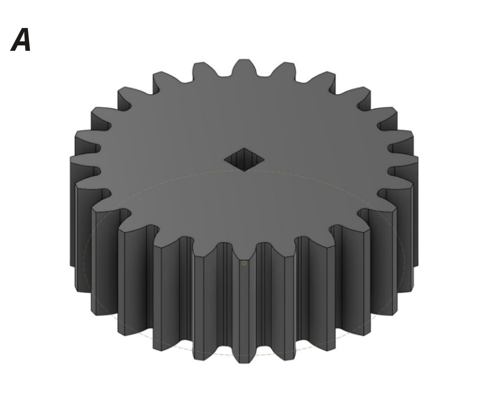
Critical Dimensions

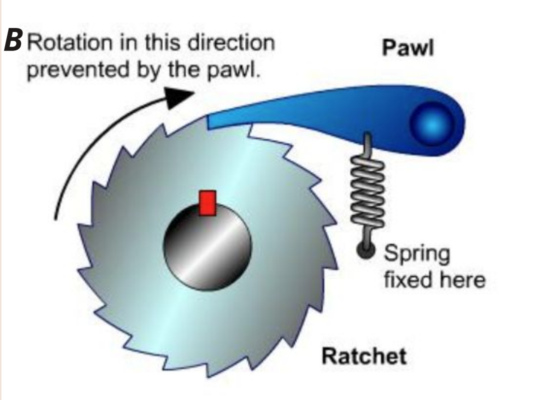
- Assuming 0.5-inch treats [1]
- Square Shaft: 0.383in x 0.383
- 0.52in Circular Pitch to allow room for treats
  - 0.52in = 3.14 * PD / 24, solving for Pitch Diameter (PD) = 4.0in
    - Pitch Diameter based on average treat size (0.5in)
- Maximum Outer Diameter: 4.65in
  - 4.0in + 8.0in/24 = **4.333in**
  - Diametral Pitch = DP = (N+2)/OD = (24+2)/4.33 = **6**
- Teeth N = 2PD/(OD-PD) = 24
- Gear Thickness 1.691in

Idea 3: Matthew Kuan

Annotated Sketch

- Define mechanical interfaces
  - 0.383in x 0.383in square hole, fitting into other geater
  - Fulfills compartment dimensions for housing and gear in dispenser
- Inner gear rotates, driving the outer gear to drop one treat at a time
- Ratcheting mechanism and recoil spring [2,3]
  - Allows lever to spring back





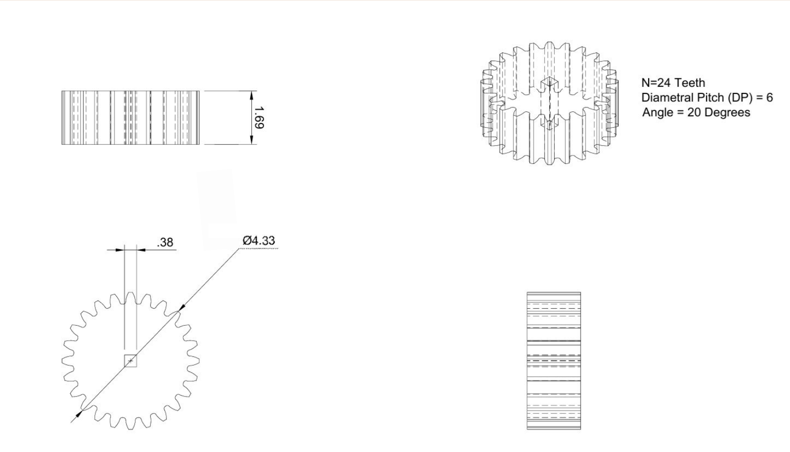


Figure: Label A Gear Design from Instructions [4].
Label B Mechanism and Recoil Mechanism to prevent rotation in opposite [3]

Figure 2: Orthographic Projection using Fusion360 Spur Gear Utility

Raffaela Mase

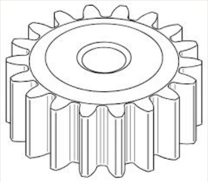
Idea 4: Raffaela Mase

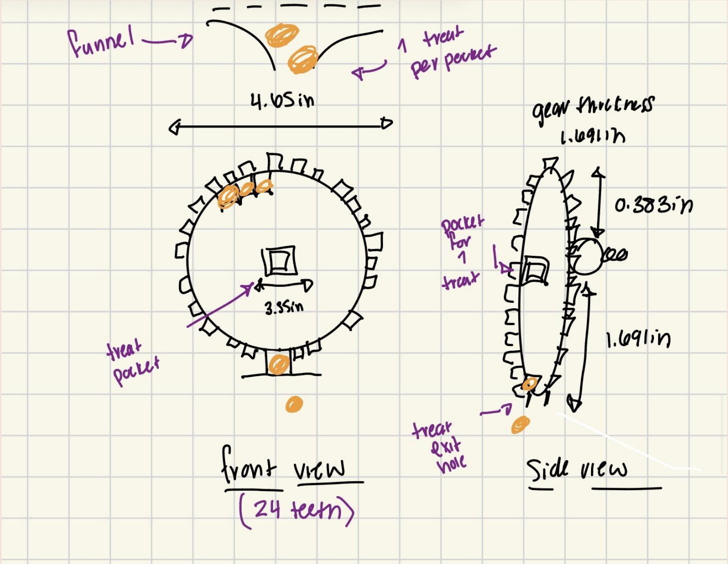
Background

People who have training dogs and use wheelchairs may have limited hand mobility. This makes it difficult to dispense treats during training. A design that releases one treat at a time with a simple push would allow for easy release and be user friendly. Our goal is to design a mechanical gear so that the user only has to push the handle once to release the treat.

How does it work?

The treat enters the dispenser from a top funnel and falls into the spaces between the gear teeth. Each space acts like a pocket to hold one treat at a time. When the user pushes the handle, the internal gear rotates the dispensing gear. As the gear rotates, the pocket holding the treat moves downward until it reaches the exit opening where the treat will fall out. A spring will then reset the handle and stop the gear at the next position so only one treat is dispensed per push.





Specific Design Ideas

Spur Gear:

- Pockets between teeth to hold individual treats
- A funnel above the gear helps treats into the gear, while a small opening at the bottom allows treat to fall out when the gear rotates.

OD: 4.65 in
Thickness: 1.691 in
Square to Connect Gear to Shaft: 0.383 x 0.383
Approx 24 teeth total

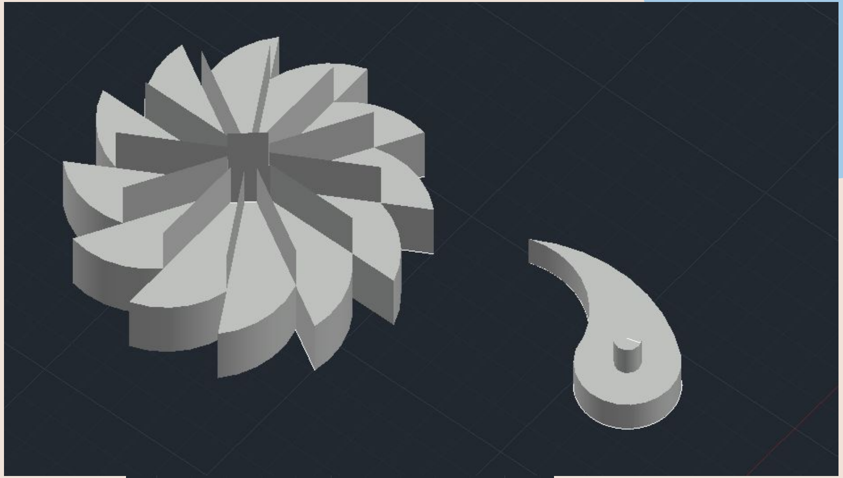
Sajib

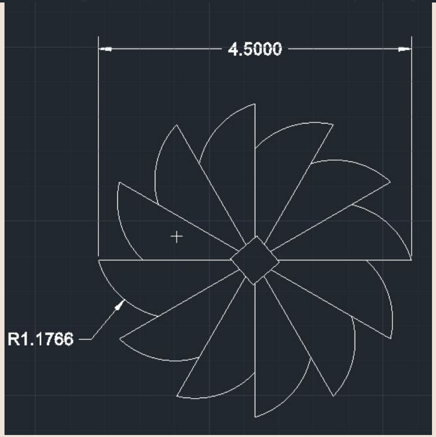
Idea 5

Concept:
Instead of having a gear that moves with the lever im think of having the lever move the pawl that forces the ratchet gear to move only with the feedback of the lever, giving the dispenser a reason to be moved back into the resting position. The gear would have capsules on the back that will hold the dog treat and once moved back the gear will dispense these treats.

Dimensions:

- *Maximum Outer Diameter*: ≤ 4.65 in.
- *Gear Thickness*: 1 in.
- *Thickness of capsule: 0.691*
- *Number of teeth:* 9
- *Diametrical Pitch: 3.010*
- *Dimension of square hole:* 0.383 in x 0.383 in





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THANK YOU Q&A

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